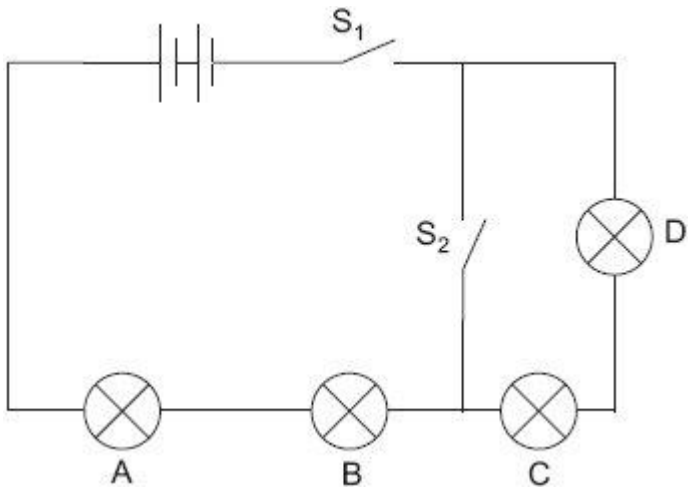
Q1. Lorna built the circuit drawn below. All the bulbs are identical.

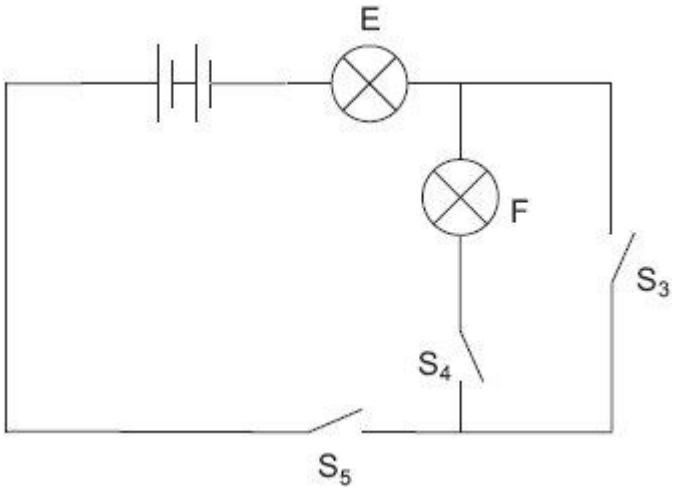


(a) Complete the table below by writing **on** or **off** for each bulb.

switch		**bulb**			
S_1	**S_2**			**A**	**B**
open	open			off	off
open	closed				
closed	open				
closed	closed				

3 marks

(b) Lorna then built a different circuit as shown below.



How could Lorna get both bulbs to light at the same time in this circuit?

.....

.....

1 mark
maximum 4 marks

Mark schemes

Q1.

(a)

A	B	C	D
off	*off*	*off*	*off*
off	off	off	off
on	on	on	on
on	on	off	off

award a mark for each correct row

3 (L6)

(b) any **one** from

- close S_4 and S_5
- only leave S_3 open

accept 'close 4 and 5'
accept 'only leave 3 open'
*'leave switch 3 **or** 3 open' is insufficient*

1 (L6)

[4]